

Each contribution answers one question with its own protocol

Contribution	Data	Protocol	Metrics	Reported in
C1 Representation	KDDTrain ⁺ UNSW train	Three-stage rule: $V \geq T$, KTA, min Q	$V(n)$, KTA(n), $Q(n)$, n^*	Figs. 4-5
C2 Entanglement	$N_{\text{train}} = 1000$ 10 runs	ZZ vs Z ablation, paired within run	ΔKTA , ΔF_1 , noise validation	Figs. 6-7
C3 Distribution shift	KDDTest ⁺ , KDDTest ⁻²¹	Prior shift, perturbation, attack composition	ΔF_1 , Holm, McNemar	Fig. 8
C4 Sample complexity	Nested subsets $N = 10^2\text{-}10^4$	Two sampling regimes, two tuning arms	Learning curve, crossover, rare F_1	Figs. 9, 11